

AUTOMATED PAYMENT REMINDER SYSTEM

Workflow Automation Project using Zapier

Project Report

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1. Executive Summary

This project presents an automated payment reminder system developed using Zapier, Google Sheets and Gmail. The purpose of the system is to reduce repetitive manual work involved in identifying pending invoices and communicating payment reminders to the relevant recipients.

The workflow operates on a scheduled basis. It retrieves invoice records from the SCIT Invoice spreadsheet, evaluates the payment status of each record, and processes only those records that are marked as Pending and contain a valid designated recipient email address. A personalized email is then generated using information from the spreadsheet. After the email is sent, the corresponding spreadsheet row is updated to record that the first reminder has been issued.

The project demonstrates the application of no-code automation to a practical business process and establishes a foundation for future features such as second reminders, payment confirmation through email replies, and escalation of overdue payments.

2. Problem Statement

Payment follow-up is often handled manually by reviewing invoice records, identifying outstanding payments, preparing reminder messages and maintaining a record of communication. When this process is repeated regularly, it can consume administrative time and introduce the possibility of missed or inconsistent follow-ups.

The project addresses this problem by creating a scheduled automation that connects invoice data with email communication. Instead of manually checking each invoice, the workflow automatically evaluates the records and performs the required reminder action when the defined conditions are satisfied.

3. Project Objectives

- Automate the identification of pending payment records from a spreadsheet.
- Execute the workflow automatically on a scheduled basis.
- Apply conditional logic so that only eligible records are processed.
- Generate personalized payment reminder emails using dynamic spreadsheet fields.
- Send the reminder from the designated Gmail account to the test recipient stored in the record.
- Update the corresponding spreadsheet row after the reminder is sent.
- Maintain a basic record of reminder activity through Reminder Count and timestamp information.
- Provide a scalable foundation for second reminders, confirmation and escalation.

4. Tools and Technologies Used

Tool	Role in Project
Zapier	Workflow automation platform used to connect the schedule, spreadsheet, filtering and Gmail actions.
Google Sheets	Stores the invoice/payment records and receives updates after reminder processing.
Gmail	Sends personalized payment reminder emails.
Google Drive	Provides the cloud-based environment in which the spreadsheet data is maintained.
CSV / Excel	Represents the underlying tabular payment data structure used for the project.

5. System Workflow

The implemented workflow follows a simple sequence of trigger, data retrieval, conditional processing, email communication and record update.

Step	Zapier Component	Function
1	Schedule by Zapier	Triggers the automation at the configured daily time.
2	Google Sheets – Get Many Spreadsheet Rows	Retrieves the invoice records from the SCIT Invoice worksheet.
3	Filter by Zapier	Allows records to continue only when Status = Pending and an Email value exists.
4	Gmail – Send Email	Sends a personalized payment reminder using dynamic invoice fields.
5	Google Sheets – Update Spreadsheet Row	Updates the processed record, including Reminder Count and optional timestamp.

Workflow Summary: Schedule → Retrieve Invoice Rows → Filter Eligible Records → Send Reminder → Update Spreadsheet

6. Input Data Structure

The SCIT Invoice worksheet acts as the primary data source. The fields used by the automation represent invoice details, payment status, reminder tracking and recipient information.

Field	Purpose
Sr	Serial/reference number
Invoice ID	Unique identifier of the invoice
Type	Payment or invoice type
Name	Name used to personalize the email
Amount	Invoice/payment amount
Account No	Bank account reference
Bank Name	Bank associated with the record
Status	Current payment status
Transaction ID	Transaction reference when available
Reminder Count	Number of reminders already issued
Escalated	Indicates whether the record has been escalated
Email	Designated test recipient address

7. Conditional Logic

The first reminder workflow uses the following eligibility rule:

Status = Pending AND Email exists

If the condition is satisfied, the record continues to the Gmail action. If the Status is Paid, Completed or another non-pending value, the record is skipped. A Pending record without a recipient email is also skipped. This prevents unintended or incomplete email actions.

8. Payment Reminder Email

The email uses dynamic fields from the spreadsheet so that each eligible invoice receives a personalized message.

Subject: Payment Reminder - Invoice [Invoice ID] Due

Dear [Name],

This is a friendly reminder that your payment for Invoice ID: [Invoice ID] of Amount: ₹[Amount] is currently pending.

Please find your payment details below for reference:

Account No: [Account No]

Bank Name: [Bank Name]

Payment Type: [Type]

Transaction ID: [Transaction ID]

Kindly ensure the payment is completed at the earliest to avoid any inconvenience.

If you have already made the payment, please disregard this message.

Thank you for your cooperation.

Regards,

Payments Team

9. Spreadsheet Update and Tracking

Once the reminder email is successfully sent, the automation updates the corresponding spreadsheet row. For the first reminder, Reminder Count changes from 0 to 1. A Sent Timestamp field may also be maintained to record when the reminder was processed. The Status field is not changed merely because a reminder was sent; it remains Pending until payment confirmation.

Stage	Reminder Tracking	Status
Before reminder	Reminder Count = 0	Pending
After first reminder	Reminder Count = 1	Pending
After payment confirmation	Future workflow	Completed

10. Testing and Expected Result

The workflow can be validated using the SCIT Invoice test data. Records with a Pending status and designated email addresses are expected to trigger the Gmail action. Records with Paid or Completed status are expected to be skipped. Pending records without an email address are also skipped.

Scenario	Expected Action
Status = Pending + Email available	Send personalized reminder
Status = Paid	Skip
Status = Completed	Skip
Status = Pending + Email blank	Skip
Successful first reminder	Update Reminder Count to 1

11. Future Enhancements

Second Payment Reminder: Send a second reminder when Status = Pending, Reminder Count = 1, Transaction ID exists and Escalated is not Yes. After sending, update Reminder Count to 2.

Payment Confirmation: Search for replies from recipients and identify messages confirming successful payment. The workflow can then update Status to Completed and record the Transaction ID.

Escalation: If payment remains pending after defined reminder stages, notify the concerned authority and mark the record as escalated.

Reporting Dashboard: Create a dashboard summarizing pending invoices, reminders sent, completed payments and escalated records.

12. Conclusion

The Automated Payment Reminder System demonstrates how a repetitive finance-related administrative process can be converted into a structured no-code workflow. By combining scheduled execution, spreadsheet-based data retrieval, conditional filtering, dynamic email generation and spreadsheet updates, the solution reduces the need for repetitive manual intervention while maintaining basic process tracking.

The current implementation focuses on the first payment reminder and establishes a clear foundation for more advanced automation. The proposed second reminder, payment confirmation and escalation features can extend the workflow into a complete payment follow-up lifecycle.

13. Project Evidence and Submission Files

The project can be documented in GitHub using the following supporting files:

- README.md – project description, workflow and implementation details.
- SCIT_Invoice_Demo.csv – sample invoice dataset used for demonstration.
- Zapier screenshots – evidence of the configured workflow, filtering and email execution.
- This report – structured documentation of the project.

Note: Designated test recipient email addresses are used for demonstration and validation of the automation. Sensitive credentials, passwords, API keys and private authentication information should not be uploaded to GitHub.